KAIZEN MATHS

Lesson starter

Matrices: Operations, Determinants and Inverses

Operations

Name _____

S1

Find $\begin{bmatrix} 0 & -4 \\ 1 & -6 \end{bmatrix} - \begin{bmatrix} -3 & -3 \\ 4 & -6 \end{bmatrix}$.

S2

Find $2\begin{bmatrix} 0 & 2 & 3 \\ 2 & 2 & 0 \\ -4 & -1 & 2 \end{bmatrix}$.

S3

Find $\begin{bmatrix} 0 & -5 & -2 \\ 3 & 6 & 4 \\ 5 & 3 & 1 \end{bmatrix} - \begin{bmatrix} 1 & -1 & 1 \\ 2 & 1 & 0 \\ -3 & -5 & -2 \end{bmatrix}$.

S4

Find $\begin{bmatrix} 0 & 7 \\ 7 & -6 \end{bmatrix} + \begin{bmatrix} -5 & -1 \\ -4 & 5 \end{bmatrix}$.

Show your working

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S1

Find $\begin{bmatrix} 0 & -4 \\ 1 & -6 \end{bmatrix} - \begin{bmatrix} -3 & -3 \\ 4 & -6 \end{bmatrix}$.

S2

Find $2\begin{bmatrix} 0 & 2 & 3 \\ 2 & 2 & 0 \\ -4 & -1 & 2 \end{bmatrix}$.

S3

Find $\begin{bmatrix} 0 & -5 & -2 \\ 3 & 6 & 4 \\ 5 & 3 & 1 \end{bmatrix} - \begin{bmatrix} 1 & -1 & 1 \\ 2 & 1 & 0 \\ -3 & -5 & -2 \end{bmatrix}$.

S4

Find $\begin{bmatrix} 0 & 7 \\ 7 & -6 \end{bmatrix} + \begin{bmatrix} -5 & -1 \\ -4 & 5 \end{bmatrix}$.

ALG-CEBD-L1 · S

Show your working

KAIZEN MATHS

Exit ticket

Matrices: Operations, Determinants and Inverses

Operations

Name _____

E1

Find $-2\begin{bmatrix} 6 & -1 & 4 \\ 5 & -3 & -5 \end{bmatrix}$.

E2

Find $\begin{bmatrix} 6 & 1 & 3 \\ -4 & 0 & 6 \\ 3 & 2 & -2 \end{bmatrix} - \begin{bmatrix} 4 & -1 & -5 \\ 0 & -3 & 3 \\ 4 & 0 & -4 \end{bmatrix}$.

E3

Find $\begin{bmatrix} -2 & -5 \\ -3 & -6 \end{bmatrix} - \begin{bmatrix} 1 & -5 \\ -3 & 8 \end{bmatrix}$.

KAIZEN MATHS

Exit ticket

Matrices: Operations, Determinants and Inverses

Operations

Name _____

E1

Find $-2\begin{bmatrix} 6 & -1 & 4 \\ 5 & -3 & -5 \end{bmatrix}$.

E2

Find $\begin{bmatrix} 6 & 1 & 3 \\ -4 & 0 & 6 \\ 3 & 2 & -2 \end{bmatrix} - \begin{bmatrix} 4 & -1 & -5 \\ 0 & -3 & 3 \\ 4 & 0 & -4 \end{bmatrix}$.

E3

Find $\begin{bmatrix} -2 & -5 \\ -3 & -6 \end{bmatrix} - \begin{bmatrix} 1 & -5 \\ -3 & 8 \end{bmatrix}$.

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Show your working

KAIZEN MATHS

Exit ticket

Matrices: Operations, Determinants and Inverses

Operations

Name _____

E1

Find $-2\begin{bmatrix} 6 & -1 & 4 \\ 5 & -3 & -5 \end{bmatrix}$.

E2

Find $\begin{bmatrix} 6 & 1 & 3 \\ -4 & 0 & 6 \\ 3 & 2 & -2 \end{bmatrix} - \begin{bmatrix} 4 & -1 & -5 \\ 0 & -3 & 3 \\ 4 & 0 & -4 \end{bmatrix}$.

E3

Find $\begin{bmatrix} -2 & -5 \\ -3 & -6 \end{bmatrix} - \begin{bmatrix} 1 & -5 \\ -3 & 8 \end{bmatrix}$.

ALG-CEBD-L1 · E

Show your working

KAIZEN MATHS

Exit ticket

Matrices: Operations, Determinants and Inverses

Operations

Name _____

E1

Find $-2\begin{bmatrix} 6 & -1 & 4 \\ 5 & -3 & -5 \end{bmatrix}$.

E2

Find $\begin{bmatrix} 6 & 1 & 3 \\ -4 & 0 & 6 \\ 3 & 2 & -2 \end{bmatrix} - \begin{bmatrix} 4 & -1 & -5 \\ 0 & -3 & 3 \\ 4 & 0 & -4 \end{bmatrix}$.

E3

Find $\begin{bmatrix} -2 & -5 \\ -3 & -6 \end{bmatrix} - \begin{bmatrix} 1 & -5 \\ -3 & 8 \end{bmatrix}$.

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Show your working

Teacher answers & working

Matrices: Operations, Determinants and Inverses · Operations

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S1

Find $\begin{bmatrix}0 & -4\\1 & -6\end{bmatrix} - \begin{bmatrix}-3 & -3\\4 & -6\end{bmatrix}$.

Answer:

$\begin{bmatrix}3 & -1\\-3 & 0\end{bmatrix}$

Working:

Check that the matrices have the same order. Both are 2 by 2, so the operation is possible.

Combine corresponding entries: top-left with top-left, top-right with top-right, and so on.

Top-left entry: 0 - -3 = 3.

Top-right entry: -4 - -3 = -1.

Bottom-left entry: 1 - 4 = -3.

Bottom-right entry: -6 - -6 = 0.

Result: $\begin{bmatrix}3 & -1\\-3 & 0\end{bmatrix}$.

S2

Find $2\begin{bmatrix}0 & 2 & 3\\2 & 2 & 0\\-4 & -1 & 2\end{bmatrix}$.

Answer:

$\begin{bmatrix}0 & 4 & 6\\4 & 4 & 0\\-8 & -2 & 4\end{bmatrix}$

Working:

Scalar multiplication means multiply every entry by the same number.

The matrix dimensions stay the same; only the entries change.

Multiply each entry by 2.

Result: $\begin{bmatrix}0 & 4 & 6\\4 & 4 & 0\\-8 & -2 & 4\end{bmatrix}$.

Teacher answers & working

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S3

Find $\begin{bmatrix} 0 & -5 & -2 \\ 3 & 6 & 4 \\ 5 & 3 & 1 \end{bmatrix} - \begin{bmatrix} 1 & -1 & 1 \\ 2 & 1 & 0 \\ -3 & -5 & -2 \end{bmatrix}$.

Answer:
$\begin{bmatrix} -1 & -4 & -3 \\ 1 & 5 & 4 \\ 8 & 8 & 3 \end{bmatrix}$

Working:

Check that both matrices are 3 by 3, so corresponding entries can be combined.

Apply the operation to each of the nine positions.

For example, first row entries become: 0 - 1 = -1, -5 - -1 = -4, and -2 - 1 = -3.

Repeat the same corresponding-entry process for the remaining rows.

Result: $\begin{bmatrix} -1 & -4 & -3 \\ 1 & 5 & 4 \\ 8 & 8 & 3 \end{bmatrix}$.

S4

Find $\begin{bmatrix} 0 & 7 \\ 7 & -6 \end{bmatrix} + \begin{bmatrix} -5 & -1 \\ -4 & 5 \end{bmatrix}$.

Answer:
$\begin{bmatrix} -5 & 6 \\ 3 & -1 \end{bmatrix}$

Working:

Check that the matrices have the same order. Both are 2 by 2, so the operation is possible.

Combine corresponding entries: top-left with top-left, top-right with top-right, and so on.

Top-left entry: 0 + -5 = -5.

Top-right entry: 7 + -1 = 6.

Bottom-left entry: 7 + -4 = 3.

Bottom-right entry: -6 + 5 = -1.

Result: $\begin{bmatrix} -5 & 6 \\ 3 & -1 \end{bmatrix}$.

Teacher answers & working

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E1

Find $-2\begin{bmatrix}6 & -1 & 4\\5 & -3 & -5\end{bmatrix}$.

Answer:

$\begin{bmatrix}-12 & 2 & -8\\-10 & 6 & 10\end{bmatrix}$

Working:

Scalar multiplication means multiply every entry by the same number.

The matrix dimensions stay the same; only the entries change.

Multiply each entry by -2.

Result: $\begin{bmatrix}-12 & 2 & -8\\-10 & 6 & 10\end{bmatrix}$.

E2

Find $\begin{bmatrix}6 & 1 & 3\\-4 & 0 & 6\\3 & 2 & -2\end{bmatrix} - \begin{bmatrix}4 & -1 & -5\\0 & -3 & 3\\4 & 0 & -4\end{bmatrix}$.

Answer:

$\begin{bmatrix}2 & 2 & 8\\-4 & 3 & 3\\-1 & 2 & 2\end{bmatrix}$

Working:

Check that both matrices are 3 by 3, so corresponding entries can be combined.

Apply the operation to each of the nine positions.

For example, first row entries become: 6 - 4 = 2, 1 - -1 = 2, and 3 - -5 = 8.

Repeat the same corresponding-entry process for the remaining rows.

Result: $\begin{bmatrix}2 & 2 & 8\\-4 & 3 & 3\\-1 & 2 & 2\end{bmatrix}$.

Teacher answers & working

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E3

Find $\begin{bmatrix} -2 & -5 \\ -3 & -6 \end{bmatrix} - \begin{bmatrix} 1 & -5 \\ -3 & 8 \end{bmatrix}$.

Answer:

$\begin{bmatrix} -3 & 0 \\ 0 & -14 \end{bmatrix}$

Working:

Check that the matrices have the same order. Both are 2 by 2, so the operation is possible.

Combine corresponding entries: top-left with top-left, top-right with top-right, and so on.

Top-left entry: -2 - 1 = -3.

Top-right entry: -5 - -5 = 0.

Bottom-left entry: -3 - -3 = 0.

Bottom-right entry: -6 - 8 = -14.

Result: $\begin{bmatrix} -3 & 0 \\ 0 & -14 \end{bmatrix}$.